

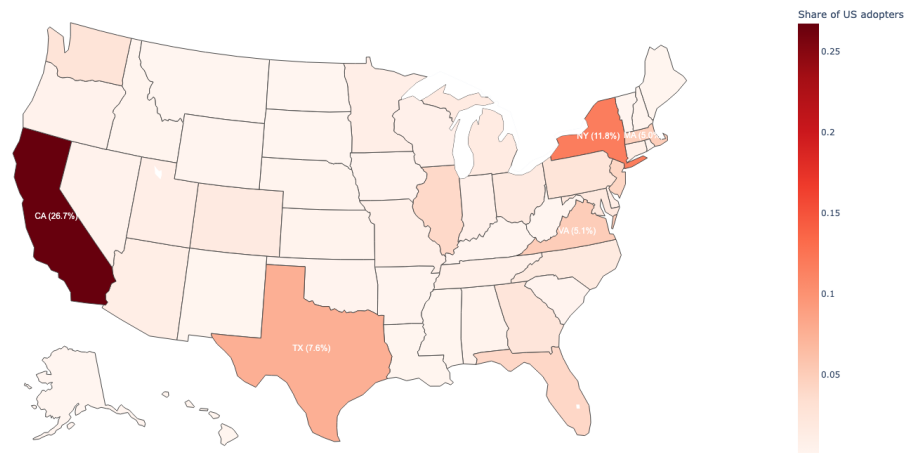


Figure A.20: Share of U.S. Adopters by State

A.17 Difference-in-Differences Estimates by Seniority

To place the descriptive patterns documented in Section 5.2 in a more formal framework, we estimate a difference-in-differences (DiD) specification, comparing employment in adopting and non-adopting firms. Specifically, we estimate the following specification separately for junior and senior workers:

$$\log(\text{Employment}_{it}) = \alpha + \sum_{j=2015Q2}^{2025Q1} \beta_j \mathbf{1}\{t = j\} \times \text{Adopt}_i + \delta_t + \text{Adopt}_i + \varepsilon_{it}, \quad (34)$$

where the dependent variable $\log(\text{Employment}_{it})$ denotes the log employment of junior (or senior) workers at firm i in quarter t .³⁷ The term $\mathbf{1}\{t = j\}$ is an indicator function that equals one if $t = j$ and zero otherwise, so that the coefficients β_j capture the differential evolution of employment for adopters relative to non-adopters in each period j . The variable Adopt_i is a dummy equal to one for firms that adopt GenAI.³⁸ Time fixed effects δ_t absorb aggregate shocks common to all firms, while Adopt_i controls for time-invariant differences between adopters and non-adopters. The error term ε_{it} captures unobserved idiosyncratic determinants of employment.

Figure A.21 reports the estimated coefficients β_j . For junior workers, the coefficients are flat and indistinguishable from zero through 2022Q4, consistent with parallel pre-trends. Starting in 2023Q1, they turn sharply negative, indicating that junior employment in adopting firms fell by about 9 percent relative to controls six quarters after the diffusion of GenAI. By contrast, coefficients for senior workers show a persistent upward trajectory throughout the sample, suggesting that adopting firms expanded senior employment more strongly than non-adopters over the last decade.

³⁷For computational efficiency we aggregate our panel data to quarterly frequency in all analyses.

³⁸Note that Adopt_i is time-invariant: a firm is defined as an adopter if it posted at least one GenAI integrator vacancy at any point during the sample (see Section 4.2 for more details). In Section 5.3, we relax this definition by exploiting variation in the timing of adoption across firms using a staggered event-study design.

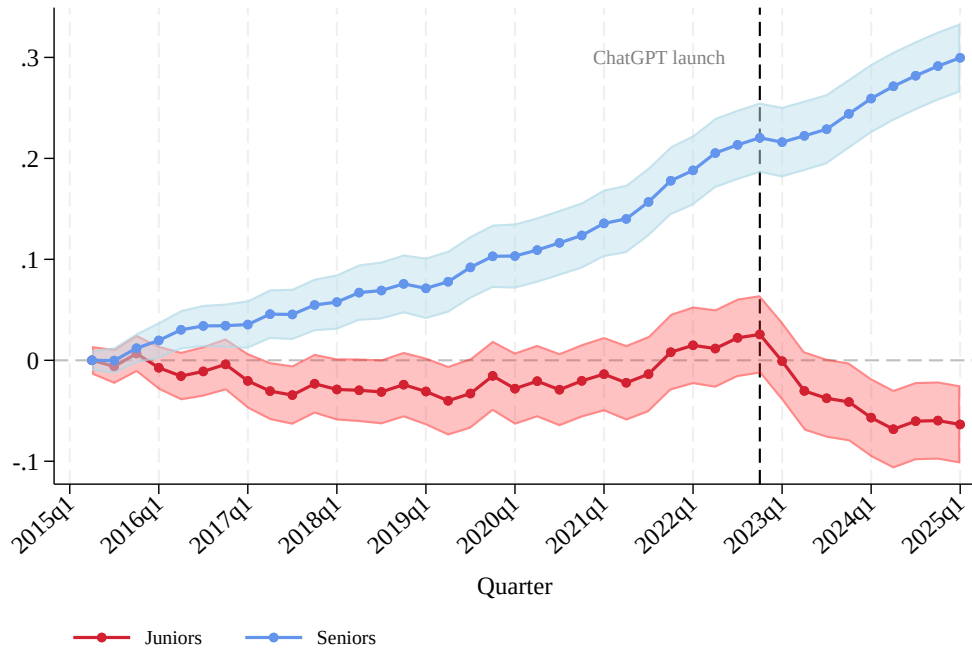


Figure A.21: Difference-in-Differences Estimates by Seniority

Notes: The figure presents the estimated DiD coefficients β_j from Equation 34, estimated separately for juniors and seniors. The adoption measure used in this analysis is time-invariant; hence, the figure compares adopters and non-adopters. For results using a time-varying adoption measure, see Section 5.3. Standard errors are clustered at the firm level.

A.18 Triple-Diff Without Industry-by-Time-by-Seniority Fixed Effects

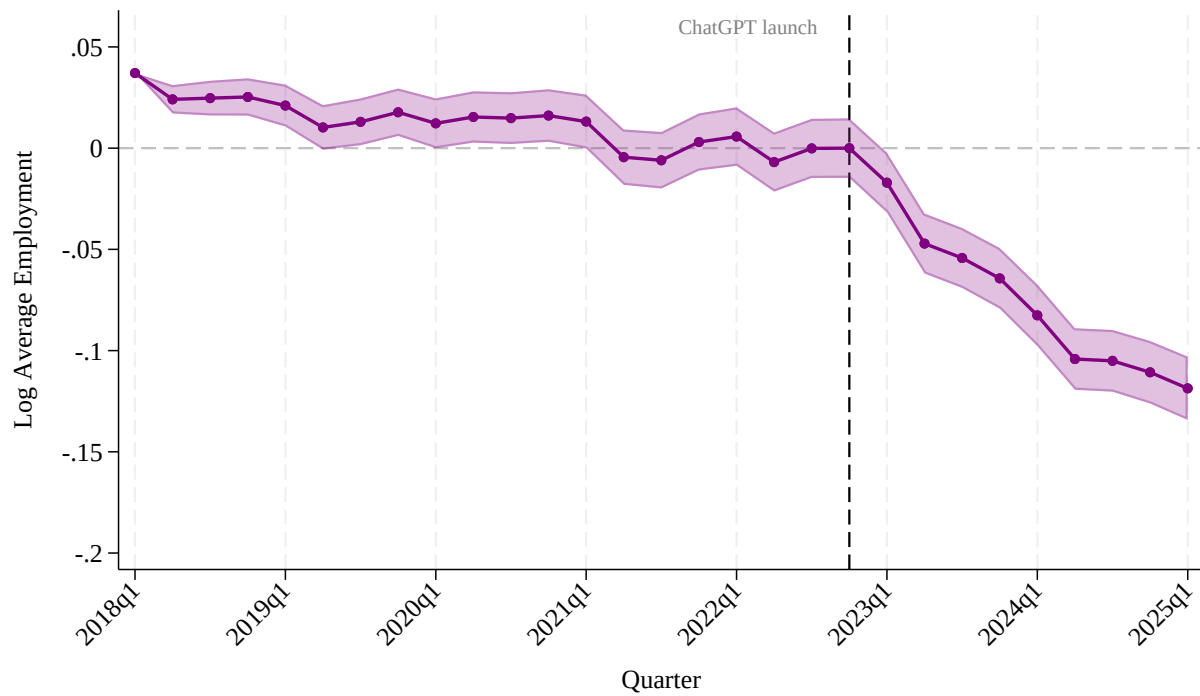


Figure A.22: Triple Differences (Without Industry-by-Time-by-Seniority Fixed Effects)

Notes: This graph shows the results of the same exercise as in Figure 5, excluding the industry-by-time-by-seniority fixed effects.